



PTT PHENOL COMPANY  
LIMITED

## PROCESS SPECIFICATION

### LEVEL INSTRUMENT PROCESS DATA SHEET

Docu. No. : 14780-8120-PS-0033  
Project No. : 120117  
Rev. No. : Z1  
Date : MAY 10, 2016  
Page : 1 / 22

TITLE :

## LEVEL INSTRUMENT PROCESS DATA SHEET

(CDN UNIT)

DISCIPLINE : PROCESS  
DOCUMENT NO. : 14780-8120-PS-0033  
PROJECT NAME : PTT Phenol Train II Project  
PROJECT NO. : 120117  
OWNER : PTT Phenol Company Limited (PPCL)

| Z1       | MAY 10, 2016      | AS BUILT            | S.W.Park     | J.S.Chung | -            | S.W.Lee  |
|----------|-------------------|---------------------|--------------|-----------|--------------|----------|
| F1       | JUN 25, 2014      | FINAL ISSUE         | S.W.Park     | J.S.Chung | -            | S.W.Lee  |
| A1       | APR 18, 2014      | APPROVED FOR DESIGN | S.W.Park     | J.S.Chung | -            | S.W.Lee  |
| O1       | DEC 30, 2013      | COMMENT ISSUE       | S.W.Park     | J.S.Chung | -            | S.W.Lee  |
| REV. NO. | DATE              | DESCRIPTION         | PREP'N       | REVIEW    | QA REVIEW    | APPROVAL |
| REV. NO. | PREPARATION DEPT. | PREPARATION         | REVIEW       | QA REVIEW | APPROVAL     |          |
| Z1       | PROCESS           | S.W.Park            | J.S.Chung    | -         | S.W.Lee      |          |
|          |                   | MAY 10, 2016        | MAY 10, 2016 | -         | MAY 10, 2016 |          |

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## PROJECT SPECIFICATION

SHEET 3 / 22

### LEVEL INSTRUMENT

| REV | DATE | BY | APVD | REV | DATE | BY | APVD |
|-----|------|----|------|-----|------|----|------|
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Rev

### Concentration, Decomposition, and Neutralization Section

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Other specifications  
referenced in this  
specification:

|  |
|--|
|  |
|  |
|  |

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Indication

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## INSTRUMENT GENERAL REQUIREMENTS

### PROJECT SPECIFICATION

963766 — 601

| REV | DATE    | BY  | APVD | REV | DATE | BY | APVD |
|-----|---------|-----|------|-----|------|----|------|
| 0   | 25JUN12 | MVD | MW   |     |      |    |      |
|     |         |     |      |     |      |    |      |
|     |         |     |      |     |      |    |      |
|     |         |     |      |     |      |    |      |

### Phenol Concentration, Decomposition, and Neutralization Section

This specification shall be supplementary to all Project Specifications, 602 - 699, issued under this project number. Unless specified otherwise on the Project Specification, the following shall apply:

#### 1 Climatic Ambient Temperature Range

The climatic ambient temperature range for field equipment design is 15 to 40°C.

#### 2 Electrical Classification

The area electrical classification shall be established by the Licensee/Contractor in accordance with local electrical codes. The Contractor shall advise all equipment vendors of the area electrical classification.

#### 3 Flanges

3.1 Carbon steel flanges are based on Material Group 1.1 of ASME B16.5 and ASME B16.47. Flange classes may differ when instrument flanges and the mating flanges on piping or equipment are specified with dissimilar materials. The Contractor shall verify that all mating flanges are properly matched with the specified instrument flanges.

3.2 Flange classes are specified in accordance with ASME B16.5 or ASME B16.47. Flange classes listed in the UOP Project Specifications are based upon design pressure and temperature conditions only and do not account for other loads. The final design of all flanges shall account for gasket seating and external loads. Differential thermal expansion, including joints between dissimilar materials, and transient thermal conditions such as start-up/shutdown and operational upset shall be accommodated.

3.3 Flanges intended for use with spiral wound gaskets shall have a flange surface finish of 125 microinch Ra minimum to 250 microinch Ra maximum. Flanges intended for use with other gaskets shall have a flange finish within the optimal range for the specified gasket. Finishes shall be judged by visual comparison with surface finish roughness standards conforming to ASME B46.1. Flange finishes shall be protected from damage during fabrication, heat treatment, shipping, storage, and installation.

#### 4 Tropical Environment

All equipment shall be tropicalized to eliminate mildew, fungi, and other detrimental effects of a tropical environment. Packaging shall be suitable for shipment and storage under tropical conditions. Packaging and "tropicalization" subject to approval by the licensee.

#### 5 Marine Environment

All field equipment shall be suitable for operation in a corrosive, salt laden, marine atmosphere.

#### 6 Tropical and Marine Environment

Transmitters and all equipment/instruments containing electrical parts shall be installed under rain shield.

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## INSTRUMENT GENERAL REQUIREMENTS

### PROJECT SPECIFICATION

963766 — 601

| REV | DATE    | BY  | APVD | REV | DATE | BY | APVD |
|-----|---------|-----|------|-----|------|----|------|
| 0   | 25JUN12 | MVD | MW   |     |      |    |      |
|     |         |     |      |     |      |    |      |
|     |         |     |      |     |      |    |      |
|     |         |     |      |     |      |    |      |

### Phenol Concentration, Decomposition, and Neutralization Section

#### 7 Fieldbus Transmitters

This specification is based on conventional (4 – 20 mA output) transmitters. UOP does not require the use of fieldbus transmitters and, if installed incorrectly, the fieldbus system can be detrimental to the control and operation of the unit. The implementation of these fieldbus transmitters is the Contractor's responsibility.

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## DIFFERENTIAL PRESSURE INSTRUMENTS

### PROJECT SPECIFICATION

963766 — 604-G

| REV | DATE    | BY  | APVD | REV | DATE | BY | APVD |
|-----|---------|-----|------|-----|------|----|------|
| 0   | 25JUN12 | MVD | MW   |     |      |    |      |
|     |         |     |      |     |      |    |      |
|     |         |     |      |     |      |    |      |
|     |         |     |      |     |      |    |      |

### GENERAL REQUIREMENTS AND INFORMATION

#### 1 Standard Drawings Referenced

UOP Standard Drawings as indicated below are part of this specification.

|       |   |       |       |               |       |       |
|-------|---|-------|-------|---------------|-------|-------|
| 6-115 | X | 6-116 | 6-120 | 6-122 + 6-123 | 6-139 | 6-140 |
| 6-142 | X | 6-144 | 6-147 | 6-148         | 6-159 | 6-161 |
|       |   |       |       |               |       | 6-162 |

#### 2 Sensor Range

The Sensing Element Lower Range Limit and Upper Range Limit are to be used as a guide only. Transmitters that have greater range capabilities are acceptable.

#### 3 Output Indicator

When an Output Indicator is specified, an integrally mounted indicator is acceptable, provided the indicator is readable from the associated control valve assembly (or other valve as denoted on the P&ID). If the piping layout is such that an integrally mounted indicator is not readable from the associated valve, then a separate indicator suitable for remote mounting shall be supplied.

#### 4 Process Temperature

The temperature shown is the *process* temperature and is not normally the operating temperature of the transmitter.

Other specifications  
referenced in this  
specification:

UOP Standard Specification: 6-18  
UOP Standard Drawing: See item 1 above  
UOP Project Specification: 601, this project



DIFFERENTIAL PRESSURE INSTRUMENTS

PROJECT SPECIFICATION

963766 — 604

| REV | DATE    | BY  | APVD | REV | DATE | BY | APVD |
|-----|---------|-----|------|-----|------|----|------|
| 0   | 25JUN12 | MVD | MW   |     |      |    |      |
|     |         |     |      |     |      |    |      |
|     |         |     |      |     |      |    |      |
|     |         |     |      |     |      |    |      |

|                             |                                                                                   |                                                  |                   |                     |                       |                                                                           |                                                                    |  |  |  |  |
|-----------------------------|-----------------------------------------------------------------------------------|--------------------------------------------------|-------------------|---------------------|-----------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------|--|--|--|--|
|                             | Item Number: 1                                                                    | Quantity: 21                                     | Data Page: 1 of 3 |                     |                       |                                                                           |                                                                    |  |  |  |  |
| BODY                        | Body/Flange Material: 316 stainless steel ‡                                       |                                                  |                   |                     | HIGH SIDE REMOTE SEAL | Process Conn Size:                      Type:                      Class: |                                                                    |  |  |  |  |
|                             | Static Pressure Rating: 105 kg/cm2(g)                                             |                                                  |                   |                     |                       | Flange Facing:                      Flange Face Finish:                   |                                                                    |  |  |  |  |
|                             | Process Connections Size: 1/2 inch                                                |                                                  |                   |                     |                       | Seal Type:                                                                |                                                                    |  |  |  |  |
|                             | Process Connections Material: 316 stainless steel ‡                               |                                                  |                   |                     |                       | Fill Fluid:                                                               |                                                                    |  |  |  |  |
|                             | Drains/Vents Material: 316 stainless steel ‡                                      |                                                  |                   |                     |                       | Diaphragm Material:                                                       |                                                                    |  |  |  |  |
|                             | Gasket/O-Ring Material: Man. Recommendation                                       |                                                  |                   |                     |                       | Mounting Flange Material:                                                 |                                                                    |  |  |  |  |
|                             | Gasket Type: Manufacturer's Recommendation                                        |                                                  |                   |                     |                       | Non-Wetted Housing Material:                                              |                                                                    |  |  |  |  |
|                             | Bolting/Nut Material: Manufacturer's Recommendation                               |                                                  |                   |                     |                       | Capillary Length:                      Armor Material:                    |                                                                    |  |  |  |  |
| TRANSMITTER SENSING ELEMENT | Diaphragm/Wetted Material: 316 stainless steel ‡                                  |                                                  |                   |                     | LOW SIDE REMOTE SEAL  | Diaphragm Extension Length:                                               |                                                                    |  |  |  |  |
|                             | Non-Wetted Material: Manufacturer's Recommendation                                |                                                  |                   |                     |                       | Process Conn Size:                      Type:                      Class: |                                                                    |  |  |  |  |
|                             | Zero Elevation/Suppression: Provide                                               |                                                  |                   |                     |                       | Flange Facing:                      Flange Face Finish:                   |                                                                    |  |  |  |  |
|                             | Sensor Lower Range Limit: 60 mBar                                                 |                                                  |                   |                     |                       | Seal Type:                                                                |                                                                    |  |  |  |  |
|                             | Sensor Upper Range Limit: 370 mBar                                                |                                                  |                   |                     |                       | Fill Fluid:                                                               |                                                                    |  |  |  |  |
|                             | Fill Fluid: Silicone                                                              |                                                  |                   |                     |                       | Diaphragm Material:                                                       |                                                                    |  |  |  |  |
| INDICATOR / CONTROLLER      | Transmitter Output: 4 - 20 mA                                                     |                                                  |                   |                     | ACCESSORIES           | Mounting Flange Material:                                                 |                                                                    |  |  |  |  |
|                             | Adjustable Dampening:                                                             |                                                  |                   |                     |                       | Non-Wetted Housing Material:                                              |                                                                    |  |  |  |  |
|                             | Housing Material: Manufacturer's Recommendation                                   |                                                  |                   |                     |                       | Capillary Length:                      Armor Material:                    |                                                                    |  |  |  |  |
|                             | Manufacturer: Honeywell, Rosemount (or equal)                                     |                                                  |                   |                     |                       | Diaphragm Extension Length:                                               |                                                                    |  |  |  |  |
| REMARKS                     | Function:                                                                         |                                                  |                   |                     |                       | Manifold Type:                                                            |                                                                    |  |  |  |  |
|                             | Controller Type:                                                                  |                                                  |                   |                     |                       | Manifold Material:                                                        |                                                                    |  |  |  |  |
|                             | Controller Action:                                                                |                                                  |                   |                     |                       | Manifold Manufacturer:                                                    |                                                                    |  |  |  |  |
|                             | Controller Output:                                                                |                                                  |                   |                     |                       | Output Indicator: See Remark §                                            |                                                                    |  |  |  |  |
| DISTRIBUTION                | Manufacturer:                                                                     |                                                  |                   |                     |                       | Air Filter/Regulator Quantity:                                            |                                                                    |  |  |  |  |
|                             | § 0 - 10 Sq. Rt. for flow; 0 - 100 linear for level and pressure differential.    |                                                  |                   |                     |                       |                                                                           |                                                                    |  |  |  |  |
|                             | ‡ All transmitter parts in contact with the process shall be 316 stainless steel. |                                                  |                   |                     |                       |                                                                           |                                                                    |  |  |  |  |
|                             | DP : Design Pressure                                                              |                                                  |                   |                     |                       |                                                                           |                                                                    |  |  |  |  |
|                             |                                                                                   |                                                  |                   |                     |                       | DT : Design Temperature                                                   |                                                                    |  |  |  |  |
| DISTRIBUTION                | Tag Number                                                                        | Service                                          | Calibrated Range  | Process Temperature | Process Pressure      | Installation                                                              | Comments                                                           |  |  |  |  |
|                             | LT-23-0803 0401D                                                                  | Preflash and Flash Columns Condenser Liquid Boot | 0 to 207 mBar     | 38 °C               | -1.0 kg/cm2(g)        | Sketch "A"                                                                | Impulse piping and fittings, shall be 316 stainless steel. Note 17 |  |  |  |  |
|                             | LT-23-2002                                                                        | Acid Aromatics Knockout Drum                     | 0 to 164 mBa      | 38 °C               | 0.01 kg/cm2(g)        | Sketch "A"                                                                | Impulse piping and fittings, shall be 316 stainless steel. Note 16 |  |  |  |  |
|                             |                                                                                   |                                                  |                   |                     |                       |                                                                           | DP : 4.0 kg/cm2(g), F.V / DT : 250 degC                            |  |  |  |  |
|                             |                                                                                   |                                                  |                   |                     |                       |                                                                           |                                                                    |  |  |  |  |
|                             |                                                                                   |                                                  |                   |                     |                       |                                                                           |                                                                    |  |  |  |  |

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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DIFFERENTIAL PRESSURE INSTRUMENTS

PROJECT SPECIFICATION

963766 — 604

| REV | DATE    | BY  | APVD | REV | DATE | BY | APVD |
|-----|---------|-----|------|-----|------|----|------|
| 0   | 25JUN12 | MVD | MW   |     |      |    |      |
|     |         |     |      |     |      |    |      |
|     |         |     |      |     |      |    |      |
|     |         |     |      |     |      |    |      |

|                             |                                                                                |                                  |                   |                     |                       |                                                                           |                                    |
|-----------------------------|--------------------------------------------------------------------------------|----------------------------------|-------------------|---------------------|-----------------------|---------------------------------------------------------------------------|------------------------------------|
|                             | Item Number: 2                                                                 | Quantity: 1                      | Data Page: 1 of 1 |                     |                       |                                                                           |                                    |
| BODY                        | Body/Flange Material: Cadmium plated CS or 316ss                               |                                  |                   |                     | HIGH SIDE REMOTE SEAL | Process Conn Size:                      Type:                      Class: |                                    |
|                             | Static Pressure Rating: 105 kg/cm2(g)                                          |                                  |                   |                     |                       | Flange Facing:                      Flange Face Finish:                   |                                    |
|                             | Process Connections Size: 1/2 inch                                             |                                  |                   |                     |                       | Seal Type:                                                                |                                    |
|                             | Process Connections Material: Cadmium plated CS or 316ss                       |                                  |                   |                     |                       | Fill Fluid:                                                               |                                    |
|                             | Drains/Vents Material: 316 stainless steel                                     |                                  |                   |                     |                       | Diaphragm Material:                                                       |                                    |
|                             | Gasket/O-Ring Material: Man. Recommendation                                    |                                  |                   |                     |                       | Mounting Flange Material:                                                 |                                    |
|                             | Gasket Type: Manufacturer's Recommendation                                     |                                  |                   |                     |                       | Non-Wetted Housing Material:                                              |                                    |
|                             | Bolting/Nut Material: Manufacturer's Recommendation                            |                                  |                   |                     |                       | Capillary Length:                      Armor Material:                    |                                    |
| TRANSMITTER SENSING ELEMENT | Diaphragm/Wetted Material: 316 stainless steel                                 |                                  |                   |                     | LOW SIDE REMOTE SEAL  | Diaphragm Extension Length:                                               |                                    |
|                             | Non-Wetted Material: Manufacturer's Recommendation                             |                                  |                   |                     |                       | Process Conn Size:                      Type:                      Class: |                                    |
|                             | Zero Elevation/Suppression: Provide                                            |                                  |                   |                     |                       | Flange Facing:                      Flange Face Finish:                   |                                    |
|                             | Sensor Lower Range Limit: 12 mBar                                              |                                  |                   |                     |                       | Seal Type:                                                                |                                    |
|                             | Sensor Upper Range Limit: 74 mBar                                              |                                  |                   |                     |                       | Fill Fluid:                                                               |                                    |
| INDICATOR / CONTROLLER      | Fill Fluid: Silicone                                                           |                                  |                   |                     | ACCESSORIES           | Diaphragm Material:                                                       |                                    |
|                             | Transmitter Output: 4 - 20 mA                                                  |                                  |                   |                     |                       | Mounting Flange Material:                                                 |                                    |
|                             | Adjustable Dampening:                                                          |                                  |                   |                     |                       | Non-Wetted Housing Material:                                              |                                    |
|                             | Housing Material: Manufacturer's Recommendation                                |                                  |                   |                     |                       | Capillary Length:                      Armor Material:                    |                                    |
| REMARKS                     | Manufacturer: Honeywell, Rosemount (or equal)                                  |                                  |                   |                     |                       | Diaphragm Extension Length:                                               |                                    |
|                             | Function:                                                                      |                                  |                   |                     |                       | Manifold Type:                                                            |                                    |
|                             | Controller Type:                                                               |                                  |                   |                     |                       | Manifold Material:                                                        |                                    |
|                             | Controller Action:                                                             |                                  |                   |                     |                       | Manifold Manufacturer:                                                    |                                    |
| DISTRIBUTION                | Controller Output:                                                             |                                  |                   |                     |                       | Output Indicator: See Remark §                                            |                                    |
|                             | Manufacturer:                                                                  |                                  |                   |                     |                       | Air Filter/Regulator Quantity:                                            |                                    |
|                             | § 0 - 10 Sq. Rt. for flow; 0 - 100 linear for level and pressure differential. |                                  |                   |                     |                       |                                                                           |                                    |
|                             | DP : Design Pressure                                                           |                                  |                   |                     |                       |                                                                           |                                    |
|                             | DT : Design Temperature                                                        |                                  |                   |                     |                       |                                                                           |                                    |
|                             |                                                                                |                                  |                   |                     |                       |                                                                           |                                    |
| DISTRIBUTION                | Tag Number                                                                     | Service                          | Calibrated Range  | Process Temperature | Process Pressure      | Installation                                                              | Comments                           |
|                             | LT-23-0601                                                                     | Concentration Cumene Quench Drum | 0 to 32 mBar      | 46 °C               | 0.07 kg/cm2(g)        | Sketch "A"                                                                | Note 6                             |
|                             |                                                                                |                                  |                   |                     |                       |                                                                           | DP : 3.5 kg/cm2(g) / DT : 120 degC |
|                             |                                                                                |                                  |                   |                     |                       |                                                                           |                                    |
|                             |                                                                                |                                  |                   |                     |                       |                                                                           |                                    |
|                             |                                                                                |                                  |                   |                     |                       |                                                                           |                                    |

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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## DIFFERENTIAL PRESSURE INSTRUMENTS

### PROJECT SPECIFICATION

963766 — 604

| REV | DATE    | BY  | APVD | REV | DATE | BY | APVD |
|-----|---------|-----|------|-----|------|----|------|
| 0   | 25JUN12 | MVD | MW   |     |      |    |      |
| 1   | 28AUG12 | MVD | MW   |     |      |    |      |
|     |         |     |      |     |      |    |      |
|     |         |     |      |     |      |    |      |

|                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                              |                  |                       |                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                           |          |  |
|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|----------|--|
| 1                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Item Number: 4                               |                  | Quantity: 11          |                                                                                                                                                                                                                                                                                                                                                        | Data Page: 1 of 2                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                           |          |  |
| BODY                                                                                                                                         | Body/Flange Material: <b>Cadmium plated CS or 316ss</b><br>Static Pressure Rating: <b>105 kg/cm2(g)</b><br>Process Connections Size: <b>1/2 inch</b><br>Process Connections Material: <b>Cadmium plated CS or 316ss</b><br>Drains/Vents Material: <b>316 stainless steel</b><br>Gasket/O-Ring Material: <b>Man. Recommendation</b><br>Gasket Type: <b>Manufacturer's Recommendation</b><br>Bolting/Nut Material: <b>Manufacturer's Recommendation</b> |                                              |                  | HIGH SIDE REMOTE SEAL | Process Conn Size:                      Type:                      Class:<br>Flange Facing:                              Flange Face Finish:<br>Seal Type:<br>Fill Fluid:<br>Diaphragm Material:<br>Mounting Flange Material:<br>Non-Wetted Housing Material:<br>Capillary Length:                      Armor Material:<br>Diaphragm Extension Length: |                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                           |          |  |
|                                                                                                                                              | Diaphragm/Wetted Material: <b>316 stainless steel</b><br>Non-Wetted Material: <b>Manufacturer's Recommendation</b><br>Zero Elevation/Suppression: <b>Provide</b><br>Sensor Lower Range Limit: <b>60 mBar</b><br>Sensor Upper Range Limit: <b>370 mBar</b><br>Fill Fluid: <b>Silicone</b>                                                                                                                                                              |                                              |                  |                       | LOW SIDE REMOTE SEAL                                                                                                                                                                                                                                                                                                                                   | Process Conn Size:                      Type:                      Class:<br>Flange Facing:                              Flange Face Finish:<br>Seal Type:<br>Fill Fluid:<br>Diaphragm Material:<br>Mounting Flange Material:<br>Non-Wetted Housing Material:<br>Capillary Length:                      Armor Material:<br>Diaphragm Extension Length: |                                                                                                                                           |          |  |
|                                                                                                                                              | Transmitter Output: <b>4 - 20 mA</b><br>Adjustable Dampening:<br>Housing Material: <b>Manufacturer's Recommendation</b><br>Manufacturer: <b>Honeywell, Rosemount (or equal)</b>                                                                                                                                                                                                                                                                       |                                              |                  |                       |                                                                                                                                                                                                                                                                                                                                                        | ACCESSORIES                                                                                                                                                                                                                                                                                                                                            | Manifold Type:<br>Manifold Material:<br>Manifold Manufacturer:<br>Output Indicator: <b>See Remark §</b><br>Air Filter/Regulator Quantity: |          |  |
|                                                                                                                                              | Function:<br>Controller Type:<br>Controller Action:<br>Controller Output:<br>Manufacturer:                                                                                                                                                                                                                                                                                                                                                            |                                              |                  |                       |                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                           |          |  |
| REMARKS<br>§ 0 - 10 Sq. Rt. for flow; 0 - 100 linear for level and pressure differential.<br>DP : Design Pressure<br>DT : Design Temperature |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                              |                  |                       |                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                           |          |  |
| DISTRIBUTION                                                                                                                                 | Tag Number                                                                                                                                                                                                                                                                                                                                                                                                                                            | Service                                      | Calibrated Range | Process Temperature   | Process Pressure                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                        | Installation                                                                                                                              | Comments |  |
|                                                                                                                                              | LT-23-0501                                                                                                                                                                                                                                                                                                                                                                                                                                            | Preflash Column Steam Heater Condensate Drum | 0 to 74 mBar     | 117 °C                | 0.9 kg/cm2(g)                                                                                                                                                                                                                                                                                                                                          | Sketch "A"                                                                                                                                                                                                                                                                                                                                             | Note 5<br>DP : 7.0 kg/cm2(g) / DT : 195 degC                                                                                              |          |  |
|                                                                                                                                              | LT-23-0701                                                                                                                                                                                                                                                                                                                                                                                                                                            | Flash Column Vaporizer Condensate Drum       | 0 to 72 mBar     | 135 °C                | 2.3 kg/cm2(g)                                                                                                                                                                                                                                                                                                                                          | Sketch "A"                                                                                                                                                                                                                                                                                                                                             | Note 7<br>DP : 7.0 kg/cm2(g) / DT : 195 degC                                                                                              |          |  |
|                                                                                                                                              | LT-23-1101                                                                                                                                                                                                                                                                                                                                                                                                                                            | Cumene Flush Drum                            | 0 to 346 mBar    | 10 °C                 | 0.7 kg/cm2(g)                                                                                                                                                                                                                                                                                                                                          | Sketch "A"                                                                                                                                                                                                                                                                                                                                             | Note 9<br>DP : 10.0 kg/cm2(g) / DT : 120 degC                                                                                             |          |  |
|                                                                                                                                              | LXT-23-1201                                                                                                                                                                                                                                                                                                                                                                                                                                           | Decomposer Feed Flush Drum                   | 0 to 115 mBar    | 46 °C                 | 0.7 kg/cm2(g)                                                                                                                                                                                                                                                                                                                                          | Sketch "A"                                                                                                                                                                                                                                                                                                                                             | Note 10<br>DP : 10.0 kg/cm2(g) / DT : 120 degC                                                                                            |          |  |
|                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                              |                  |                       |                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                           |          |  |
| ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                              |                  |                       |                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                           |          |  |

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DIFFERENTIAL PRESSURE INSTRUMENTS

PROJECT SPECIFICATION

963766 — 604

| REV | DATE    | BY  | APVD | REV | DATE | BY | APVD |
|-----|---------|-----|------|-----|------|----|------|
| 0   | 25JUN12 | MVD | MW   |     |      |    |      |
|     |         |     |      |     |      |    |      |
|     |         |     |      |     |      |    |      |
|     |         |     |      |     |      |    |      |

|                                                                                                                       |                                                                                   |                   |                  |                     |                       |                                     |                                                                    |
|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------|------------------|---------------------|-----------------------|-------------------------------------|--------------------------------------------------------------------|
| Item Number: 6                                                                                                        | Quantity: 3                                                                       | Data Page: 1 of 1 |                  |                     |                       |                                     |                                                                    |
|                                                                                                                       |                                                                                   |                   |                  |                     |                       |                                     |                                                                    |
| BODY                                                                                                                  | Body/Flange Material: 316 stainless steel ‡                                       |                   |                  |                     | HIGH SIDE REMOTE SEAL | Process Conn Size: Type: Class:     |                                                                    |
|                                                                                                                       | Static Pressure Rating: 105 kg/cm2(g)                                             |                   |                  |                     |                       | Flange Facing: Flange Face Finish:  |                                                                    |
|                                                                                                                       | Process Connections Size: 1/2 inch                                                |                   |                  |                     |                       | Seal Type:                          |                                                                    |
|                                                                                                                       | Process Connections Material: 316 stainless steel ‡                               |                   |                  |                     |                       | Fill Fluid:                         |                                                                    |
|                                                                                                                       | Drains/Vents Material: 316 stainless steel ‡                                      |                   |                  |                     |                       | Diaphragm Material:                 |                                                                    |
|                                                                                                                       | Gasket/O-Ring Material: Man. Recommendation                                       |                   |                  |                     |                       | Mounting Flange Material:           |                                                                    |
|                                                                                                                       | Gasket Type: Manufacturer's Recommendation                                        |                   |                  |                     |                       | Non-Wetted Housing Material:        |                                                                    |
|                                                                                                                       | Bolting/Nut Material: Manufacturer's Recommendation                               |                   |                  |                     |                       | Capillary Length: Armor Material:   |                                                                    |
| TRANSMITTER SENSING ELEMENT                                                                                           | Diaphragm/Wetted Material: 316 stainless steel ‡                                  |                   |                  |                     | LOW SIDE REMOTE SEAL  | Diaphragm Extension Length:         |                                                                    |
|                                                                                                                       | Non-Wetted Material: Manufacturer's Recommendation                                |                   |                  |                     |                       | Process Conn Size: Type: Class:     |                                                                    |
|                                                                                                                       | Zero Elevation/Suppression: Provide                                               |                   |                  |                     |                       | Flange Facing: Flange Face Finish:  |                                                                    |
|                                                                                                                       | Sensor Lower Range Limit: 12 mBar                                                 |                   |                  |                     |                       | Seal Type:                          |                                                                    |
|                                                                                                                       | Sensor Upper Range Limit: 74 mBar                                                 |                   |                  |                     |                       | Fill Fluid:                         |                                                                    |
| INDICATOR / CONTROLLER                                                                                                | Fill Fluid: Silicone                                                              |                   |                  |                     | ACCESSORIES           | Diaphragm Material:                 |                                                                    |
|                                                                                                                       | Transmitter Output: 4 - 20 mA                                                     |                   |                  |                     |                       | Mounting Flange Material:           |                                                                    |
|                                                                                                                       | Adjustable Dampening:                                                             |                   |                  |                     |                       | Non-Wetted Housing Material:        |                                                                    |
|                                                                                                                       | Housing Material: Manufacturer's Recommendation                                   |                   |                  |                     |                       | Capillary Length: Armor Material:   |                                                                    |
| REMARKS                                                                                                               | Manufacturer: Honeywell, Rosemount (or equal)                                     |                   |                  |                     |                       | Diaphragm Extension Length:         |                                                                    |
|                                                                                                                       | Function:                                                                         |                   |                  |                     |                       | Manifold Type:                      |                                                                    |
|                                                                                                                       | Controller Type:                                                                  |                   |                  |                     |                       | Manifold Material:                  |                                                                    |
|                                                                                                                       | Controller Action:                                                                |                   |                  |                     |                       | Manifold Manufacturer:              |                                                                    |
| DISTRIBUTION                                                                                                          | Controller Output:                                                                |                   |                  |                     |                       | Output Indicator: See Remark §      |                                                                    |
|                                                                                                                       | Manufacturer:                                                                     |                   |                  |                     |                       | Air Filter/Regulator Quantity:      |                                                                    |
|                                                                                                                       | § 0 - 10 Sq. Rt. for flow; 0 - 100 linear for level and pressure differential.    |                   |                  |                     |                       |                                     |                                                                    |
|                                                                                                                       | ‡ All transmitter parts in contact with the process shall be 316 stainless steel. |                   |                  |                     |                       |                                     |                                                                    |
| DP : Design Pressure                                                                                                  |                                                                                   |                   |                  |                     |                       |                                     |                                                                    |
| DT : Design Temperature                                                                                               |                                                                                   |                   |                  |                     |                       |                                     |                                                                    |
| DISTRIBUTION                                                                                                          | Tag Number                                                                        | Service           | Calibrated Range | Process Temperature | Process Pressure      | Installation                        | Comments                                                           |
|                                                                                                                       | LXT-23-1303                                                                       | Decomposer Drum   | 0 to 36 mBar     | 60 °C               | 0.7 kg/cm2(g)         | Sketch "A"                          | Impulse piping and fittings, shall be 316 stainless steel. Note 11 |
|                                                                                                                       |                                                                                   |                   |                  |                     |                       |                                     | DP : 11.0 kg/cm2(g) / DT : 250 degC                                |
|                                                                                                                       | LT-23-1301                                                                        | Decomposer Drum   | 0 to 72 mBar     | 60 °C               | 0.7 kg/cm2(g)         | Sketch "A"                          | Impulse piping and fittings, shall be 316 stainless steel. Note 12 |
|                                                                                                                       |                                                                                   |                   |                  |                     |                       |                                     | DP : 11.0 kg/cm2(g) / DT : 250 degC                                |
|                                                                                                                       | LXT-23-1302                                                                       | Decomposer Drum   | 0 to 36 mBar     | 60 °C               | 0.7 kg/cm2(g)         | Sketch "A"                          | Impulse piping and fittings, shall be 316 stainless steel. Note 11 |
|                                                                                                                       |                                                                                   |                   |                  |                     |                       | DP : 11.0 kg/cm2(g) / DT : 250 degC |                                                                    |
| ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION. |                                                                                   |                   |                  |                     |                       |                                     |                                                                    |

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DIFFERENTIAL PRESSURE INSTRUMENTS

PROJECT SPECIFICATION

963766 — 604

| REV | DATE    | BY  | APVD | REV | DATE | BY | APVD |
|-----|---------|-----|------|-----|------|----|------|
| 0   | 25JUN12 | MVD | MW   |     |      |    |      |
| 1   | 28AUG12 | MVD | MW   |     |      |    |      |
|     |         |     |      |     |      |    |      |
|     |         |     |      |     |      |    |      |

|                             |                                                                                                                                                                                                                                                                                                         |                        |                   |                       |                                                   |              |                                                                                                      |  |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------|-----------------------|---------------------------------------------------|--------------|------------------------------------------------------------------------------------------------------|--|
| 1                           | Item Number: 9                                                                                                                                                                                                                                                                                          | Quantity: 4            | Data Page: 1 of 1 |                       |                                                   |              |                                                                                                      |  |
| BODY                        | Body/Flange Material: Cadmium plated CS or 316ss                                                                                                                                                                                                                                                        |                        |                   | HIGH SIDE REMOTE SEAL | Process Conn Size: 3 inch Type: Flange Class: 300 |              |                                                                                                      |  |
|                             | Static Pressure Rating: 105 kg/cm2(g)                                                                                                                                                                                                                                                                   |                        |                   |                       | Flange Facing: Raised Face Flange Face Finish:    |              |                                                                                                      |  |
|                             | Process Connections Size: 1/2 inch                                                                                                                                                                                                                                                                      |                        |                   |                       | Seal Type: Wafer                                  |              |                                                                                                      |  |
|                             | Process Connections Material: Cadmium plated CS or 316ss                                                                                                                                                                                                                                                |                        |                   |                       | Fill Fluid: Silicone §                            |              |                                                                                                      |  |
|                             | Drains/Vents Material: 316 stainless steel                                                                                                                                                                                                                                                              |                        |                   |                       | Diaphragm Material: 316 stainless steel           |              |                                                                                                      |  |
|                             | Gasket/O-Ring Material: Man. Recommendation                                                                                                                                                                                                                                                             |                        |                   |                       | Mounting Flange Material:                         |              |                                                                                                      |  |
|                             | Gasket Type: Manufacturer's Recommendation                                                                                                                                                                                                                                                              |                        |                   |                       | Non-Wetted Housing Material: Mfr. Standard        |              |                                                                                                      |  |
|                             | Bolting/Nut Material: Manufacturer's Recommendation                                                                                                                                                                                                                                                     |                        |                   |                       | Capillary Length: ‡ Armor Material: Mfr. Std.     |              |                                                                                                      |  |
| TRANSMITTER SENSING ELEMENT | Diaphragm/Wetted Material: 316 stainless steel                                                                                                                                                                                                                                                          |                        |                   | LOW SIDE REMOTE SEAL  | Diaphragm Extension Length:                       |              |                                                                                                      |  |
|                             | Non-Wetted Material: Manufacturer's Recommendation                                                                                                                                                                                                                                                      |                        |                   |                       | Process Conn Size: 3 inch Type: Flange Class: 300 |              |                                                                                                      |  |
|                             | Zero Elevation/Suppression: Provide                                                                                                                                                                                                                                                                     |                        |                   |                       | Flange Facing: Raised Face Flange Face Finish:    |              |                                                                                                      |  |
|                             | Sensor Lower Range Limit: 60 mBar                                                                                                                                                                                                                                                                       |                        |                   |                       | Seal Type: Wafer                                  |              |                                                                                                      |  |
|                             | Sensor Upper Range Limit: 370 mBar                                                                                                                                                                                                                                                                      |                        |                   |                       | Fill Fluid: Silicone §                            |              |                                                                                                      |  |
| INDICATOR / CONTROLLER      | Fill Fluid: Silicone                                                                                                                                                                                                                                                                                    |                        |                   | ACCESSORIES           | Diaphragm Material: 316 stainless steel           |              |                                                                                                      |  |
|                             | Transmitter Output: 4 - 20 mA                                                                                                                                                                                                                                                                           |                        |                   |                       | Mounting Flange Material:                         |              |                                                                                                      |  |
|                             | Adjustable Dampening:                                                                                                                                                                                                                                                                                   |                        |                   |                       | Non-Wetted Housing Material: Mfr. Standard        |              |                                                                                                      |  |
|                             | Housing Material: Manufacturer's Recommendation                                                                                                                                                                                                                                                         |                        |                   |                       | Capillary Length: ‡ Armor Material: Mfr. Std.     |              |                                                                                                      |  |
| REMARKS                     | Manufacturer: Honeywell, Rosemount (or equal)                                                                                                                                                                                                                                                           |                        |                   |                       | Diaphragm Extension Length:                       |              |                                                                                                      |  |
|                             | Function:                                                                                                                                                                                                                                                                                               |                        |                   |                       | Manifold Type:                                    |              |                                                                                                      |  |
|                             | Controller Type:                                                                                                                                                                                                                                                                                        |                        |                   |                       | Manifold Material:                                |              |                                                                                                      |  |
|                             | Controller Action:                                                                                                                                                                                                                                                                                      |                        |                   |                       | Manifold Manufacturer:                            |              |                                                                                                      |  |
| DISTRIBUTION                | Controller Output:                                                                                                                                                                                                                                                                                      |                        |                   |                       | Output Indicator: 0 - 100 Linear                  |              |                                                                                                      |  |
|                             | Manufacturer:                                                                                                                                                                                                                                                                                           |                        |                   |                       | Air Filter/Regulator Quantity:                    |              |                                                                                                      |  |
|                             | § Capillary fill shall be suitable for the pressure and maximum operating temperature indicated.                                                                                                                                                                                                        |                        |                   |                       |                                                   |              |                                                                                                      |  |
|                             | ‡ The capillary length shall be of sufficient length on each leg for installation of the transmitter below the lowest tap. The capillary tubing shall be protected from direct contact with sunlight to minimize measurement errors caused by temperature differences between high and low capillaries. |                        |                   |                       |                                                   |              |                                                                                                      |  |
| DP : Design Pressure        |                                                                                                                                                                                                                                                                                                         |                        |                   |                       |                                                   |              |                                                                                                      |  |
| DT : Design Temperature     |                                                                                                                                                                                                                                                                                                         |                        |                   |                       |                                                   |              |                                                                                                      |  |
| 1                           | Tag Number                                                                                                                                                                                                                                                                                              | Service                | Calibrated Range  | Process Temperature   | Process Pressure                                  | Installation | Comments                                                                                             |  |
|                             | LT-23-0402                                                                                                                                                                                                                                                                                              | Preflash Column Bottom | 0 to 107 mBar     | 70 °C                 | -1.0 kg/cm2(g)                                    | Sketch "B"   | Capillary fill max temperature = 250°C<br>Notes 2, 3<br>DP : 3.5 kg/cm2(g), F.V / DT : 250, 195 degC |  |
|                             | LT-23-0403                                                                                                                                                                                                                                                                                              | Preflash Column Bottom | 0 to 107 mBar     | 70 °C                 | -1.0 kg/cm2(g)                                    | Sketch "B"   | Capillary fill max temperature = 250°C<br>Notes 3, 4<br>DP : 3.5 kg/cm2(g), F.V / DT : 250, 195 degC |  |
|                             | LT-23-0801                                                                                                                                                                                                                                                                                              | Flash Column Bottom    | 0 to 219 mBar     | 95 °C                 | -0.24 kg/cm2(g)                                   | Sketch "B"   | Capillary fill max temperature = 125°C<br>Notes 3, 8<br>DP : 3.5 kg/cm2(g), F.V / DT : 250, 195 degC |  |
|                             | LXT-23-0802                                                                                                                                                                                                                                                                                             | Flash Column Bottom    | 0 to 219 mBar     | 95 °C                 | -0.24 kg/cm2(g)                                   | Sketch "B"   | Capillary fill max temperature = 125°C<br>Notes 3, 8<br>DP : 3.5 kg/cm2(g), F.V / DT : 250, 195 degC |  |
|                             | CANCELED                                                                                                                                                                                                                                                                                                |                        |                   |                       |                                                   |              |                                                                                                      |  |

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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## DIFFERENTIAL PRESSURE INSTRUMENTS

### PROJECT SPECIFICATION

963766 — 604

| REV | DATE    | BY  | APVD | REV | DATE | BY | APVD |
|-----|---------|-----|------|-----|------|----|------|
| 0   | 28AUG12 | MVD | MW   |     |      |    |      |
|     |         |     |      |     |      |    |      |
|     |         |     |      |     |      |    |      |
|     |         |     |      |     |      |    |      |

|                                                                                                                       |                                                                                                                                                                         |                                   |                  |                          |                                                                        |                       |                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|------------------|--------------------------|------------------------------------------------------------------------|-----------------------|----------------------------------------------------------------------------------------------|
| Item Number: <b>10</b>                                                                                                |                                                                                                                                                                         | Quantity: <b>1</b>                |                  | Data Page: <b>1 of 1</b> |                                                                        |                       |                                                                                              |
| BODY                                                                                                                  | Body/Flange Material: <b>Cadmium plated CS or 316ss</b>                                                                                                                 |                                   |                  | HIGH SIDE REMOTE SEAL    | Process Conn Size: <b>3 inch</b> Type: <b>Flange</b> Class: <b>300</b> |                       |                                                                                              |
|                                                                                                                       | Static Pressure Rating: <b>105 kg/cm2(g)</b>                                                                                                                            |                                   |                  |                          | Flange Facing: <b>Raised Face</b> Flange Face Finish:                  |                       |                                                                                              |
|                                                                                                                       | Process Connections Size: <b>1/2 inch</b>                                                                                                                               |                                   |                  |                          | Seal Type: <b>Wafer</b>                                                |                       |                                                                                              |
|                                                                                                                       | Process Connections Material: <b>Cadmium plated CS or 316ss</b>                                                                                                         |                                   |                  |                          | Fill Fluid: <b>Silicone S</b>                                          |                       |                                                                                              |
|                                                                                                                       | Drains/Vents Material: <b>316 stainless steel</b>                                                                                                                       |                                   |                  |                          | Diaphragm Material: <b>316 stainless steel</b>                         |                       |                                                                                              |
|                                                                                                                       | Gasket/O-Ring Material: <b>Man. Recommendation</b>                                                                                                                      |                                   |                  |                          | Mounting Flange Material:                                              |                       |                                                                                              |
|                                                                                                                       | Gasket Type: <b>Manufacturer's Recommendation</b>                                                                                                                       |                                   |                  |                          | Non-Wetted Housing Material: <b>Mfr. Standard</b>                      |                       |                                                                                              |
|                                                                                                                       | Bolting/Nut Material: <b>Manufacturer's Recommendation</b>                                                                                                              |                                   |                  |                          | Capillary Length: <b>‡</b> Armor Material: <b>Mfr. Std.</b>            |                       |                                                                                              |
| TRANSMITTER SENSING ELEMENT                                                                                           | Diaphragm/Wetted Material: <b>316 stainless steel</b>                                                                                                                   |                                   |                  | LOW SIDE REMOTE SEAL     | Diaphragm Extension Length:                                            |                       |                                                                                              |
|                                                                                                                       | Non-Wetted Material: <b>Manufacturer's Recommendation</b>                                                                                                               |                                   |                  |                          | Process Conn Size: Type: Class:                                        |                       |                                                                                              |
|                                                                                                                       | Zero Elevation/Suppression: <b>Provide</b>                                                                                                                              |                                   |                  |                          | Flange Facing: Flange Face Finish:                                     |                       |                                                                                              |
|                                                                                                                       | Sensor Lower Range Limit: <b>60 mBar</b>                                                                                                                                |                                   |                  |                          | Seal Type:                                                             |                       |                                                                                              |
|                                                                                                                       | Sensor Upper Range Limit: <b>370 mBar</b>                                                                                                                               |                                   |                  |                          | Fill Fluid:                                                            |                       |                                                                                              |
| INDICATOR / CONTROLLER                                                                                                | Fill Fluid: <b>Silicone</b>                                                                                                                                             |                                   |                  | ACCESSORIES              | Diaphragm Material:                                                    |                       |                                                                                              |
|                                                                                                                       | Transmitter Output: <b>4 - 20 mA</b>                                                                                                                                    |                                   |                  |                          | Mounting Flange Material:                                              |                       |                                                                                              |
|                                                                                                                       | Adjustable Dampening:                                                                                                                                                   |                                   |                  |                          | Non-Wetted Housing Material:                                           |                       |                                                                                              |
|                                                                                                                       | Housing Material: <b>Manufacturer's Recommendation</b>                                                                                                                  |                                   |                  |                          | Capillary Length: Armor Material:                                      |                       |                                                                                              |
|                                                                                                                       | Manufacturer: <b>Honeywell, Rosemount (or equal)</b>                                                                                                                    |                                   |                  |                          | Diaphragm Extension Length:                                            |                       |                                                                                              |
| REMARKS                                                                                                               | Function:                                                                                                                                                               |                                   |                  |                          |                                                                        |                       |                                                                                              |
|                                                                                                                       | Controller Type:                                                                                                                                                        |                                   |                  |                          |                                                                        |                       |                                                                                              |
|                                                                                                                       | Controller Action:                                                                                                                                                      |                                   |                  |                          |                                                                        |                       |                                                                                              |
|                                                                                                                       | Controller Output:                                                                                                                                                      |                                   |                  |                          |                                                                        |                       |                                                                                              |
| DISTRIBUTION                                                                                                          | Manufacturer:                                                                                                                                                           |                                   |                  |                          |                                                                        |                       |                                                                                              |
|                                                                                                                       | § Capillary fill shall be suitable for the pressure and maximum operating temperature indicated.                                                                        |                                   |                  |                          |                                                                        |                       |                                                                                              |
|                                                                                                                       | ‡ The capillary length shall be determined by the contractor. The capillary tubing shall be protected from direct contact with sunlight to minimize measurement errors. |                                   |                  |                          |                                                                        |                       |                                                                                              |
|                                                                                                                       | DP : Design Pressure                                                                                                                                                    |                                   |                  |                          |                                                                        |                       |                                                                                              |
| DT : Design Temperature                                                                                               |                                                                                                                                                                         |                                   |                  |                          |                                                                        |                       |                                                                                              |
| DISTRIBUTION                                                                                                          | Tag Number                                                                                                                                                              | Service                           | Calibrated Range | Process Temperature      | Process Pressure                                                       | Installation          | Comments                                                                                     |
|                                                                                                                       | LT-23-1501                                                                                                                                                              | Decomposer Acid Injection Tote    | 0 to 168 mBar    | 30 °C                    | 0.01 kg/cm2(g)                                                         | Similar to Sketch "B" | Capillary fill max temperature = 120 °C<br>Notes 3, 18, 19                                   |
|                                                                                                                       | LT-23-1901                                                                                                                                                              | Neutralizing Agent Injection Tote | 0 to 168 mBar    | 30 °C                    | 0.01 kg/cm2(g)                                                         | Similar to Sketch "B" | DP : 3.5 kg/cm2(g) / DT : 120 degC<br>Capillary fill max temperature = 120 °C<br>Notes 3, 20 |
|                                                                                                                       |                                                                                                                                                                         |                                   |                  |                          |                                                                        |                       | DP : 3.5 kg/cm2(g) / DT : 120 degC                                                           |
|                                                                                                                       |                                                                                                                                                                         |                                   |                  |                          |                                                                        |                       |                                                                                              |
|                                                                                                                       |                                                                                                                                                                         |                                   |                  |                          |                                                                        |                       |                                                                                              |
| ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION. |                                                                                                                                                                         |                                   |                  |                          |                                                                        |                       |                                                                                              |

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## DIFFERENTIAL PRESSURE INSTRUMENTS

### PROJECT SPECIFICATION

963766 — 604-N

| REV | DATE    | BY  | APVD | REV | DATE | BY | APVD |
|-----|---------|-----|------|-----|------|----|------|
| 0   | 25JUN12 | MVD | MW   |     |      |    |      |
| 1   | 28AUG12 | MVD | MW   |     |      |    |      |
|     |         |     |      |     |      |    |      |
|     |         |     |      |     |      |    |      |

#### Notes:

- Four transmitters are piped from one orifice flange. The transmitters shall be installed in accordance with UOP Standard Drawing 6-116 and shall have individual isolation.
- The transmitter differential range specified is based upon a level range of ~~1600~~ 1200 mm and a liquid specific gravity of 0.916. Provide zero elevation as required to compensate for the static head of capillary fill fluid between the upper and lower level connections and the transmitter. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
- Impulse piping and fittings shall be 316 stainless steel.
- The transmitter differential range specified is based upon a level range of 1200 mm and a liquid specific gravity of 0.916. Provide zero elevation as required to compensate for the static head of capillary fill fluid between the upper and lower level connections and the transmitter. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
- The transmitter differential range specified is based upon a level range of 800 mm and a liquid specific gravity of 0.947. Fill the reference leg with Water and provide zero elevation as required to compensate for the reference head. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
- The transmitter differential range specified is based upon a level range of 400 mm and a liquid specific gravity of 0.841. Fill the reference leg with Cumene and provide zero elevation as required to compensate for the reference head. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
- The transmitter differential range specified is based upon a level range of 800 mm and a liquid specific gravity of 0.931. Fill the reference leg with Water and provide zero elevation as required to compensate for the reference head. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
- The transmitter differential range specified is based upon a level range of 2400 mm and a liquid specific gravity of 0.935. Provide zero elevation as required to compensate for the static head of capillary fill fluid between the upper and lower level connections and the transmitter. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
- The transmitter differential range specified is based upon a level range of 4200 mm and a liquid specific gravity of 0.841. Fill the reference leg with Cumene and provide zero elevation as required to compensate for the reference head. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.

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## DIFFERENTIAL PRESSURE INSTRUMENTS

### PROJECT SPECIFICATION

963766 — 604-N

| REV | DATE    | BY  | APVD | REV | DATE | BY | APVD |
|-----|---------|-----|------|-----|------|----|------|
| 0   | 25JUN12 | MVD | MW   |     |      |    |      |
|     |         |     |      |     |      |    |      |
|     |         |     |      |     |      |    |      |
|     |         |     |      |     |      |    |      |

#### Notes:

10. The transmitter differential range specified is based upon a level range of 1400 mm and a liquid specific gravity of 0.841. Fill the reference leg with Cumene and provide zero elevation as required to compensate for the reference head. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
11. The transmitter differential range specified is based upon a level range of 400 mm and a liquid specific gravity of 0.920. Fill the reference leg with Decomposer Drum liquid and provide zero elevation as required to compensate for the reference head. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
12. The transmitter differential range specified is based upon a level range of 800 mm and a liquid specific gravity of 0.920. Fill the reference leg with Decomposer Drum liquid and provide zero elevation as required to compensate for the reference head. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
13. Provide a first order lag signal filtering function, with adjustable time constant, for reduction of process signal noise. The signal filtering may be accomplished either by analog or digital means, at the transmitter. Alternatively, the signal filtering can be performed at the DCS. The time constant shall initially be set at the minimum value.
14. A horizontal long radius elbow in the regular piping shall be used for this meter. The pressure taps shall be located at an angle of 45 degrees from the inlet weld seam, diametrically opposite one another, at the extreme inside and outside of the bend. Provide 25 diameters of straight run pipe upstream of the long radius elbow and 10 diameters downstream. The meter manifold and meter installation shall otherwise conform to UOP Standard Drawing 6-116.
15. Two transmitters are piped from one orifice flange. The transmitters shall be installed in accordance with UOP Standard Drawing 6-116 and shall have individual isolation.
16. The transmitter differential range specified is based upon a level range of 2100 mm and a liquid specific gravity of 0.800. Fill the reference leg with Acid Aromatics Knockout Drum liquid and provide zero elevation as required to compensate for the reference head. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
17. The transmitter differential range specified is based upon a level range of 2500 mm and a liquid specific gravity of 0.848. Fill the reference leg with Preflash and Flash Columns Condenser liquid and provide zero elevation as required to compensate for the reference head. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.

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## DIFFERENTIAL PRESSURE INSTRUMENTS

### PROJECT SPECIFICATION

963766 — 604-N

| REV | DATE    | BY  | APVD | REV | DATE | BY | APVD |
|-----|---------|-----|------|-----|------|----|------|
| 0   | 25JUN12 | MVD | MW   |     |      |    |      |
| 1   | 28AUG12 | MVD | MW   |     |      |    |      |
|     |         |     |      |     |      |    |      |
|     |         |     |      |     |      |    |      |

#### Notes:

18. The transmitter differential range specified is based upon a level range of 1000 mm and a liquid specific gravity of 1.716. Provide zero elevation as required to compensate for the static head of capillary fill fluid. ~~between the upper and lower level connections and the transmitter.~~ The low pressure tap shall be vented to the atmosphere and protected from rain, wind and insects. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
19. Contractor to confirm the calibration range.
20. The transmitter differential range specified is based upon a level range of 1000 mm and a liquid specific gravity of 0.8488. Provide zero elevation as required to compensate for the static head of capillary fill fluid. The low pressure tap shall be vented to the atmosphere and protected from rain, wind and insects. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.



**posco**  
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## DIFFERENTIAL PRESSURE INSTRUMENTS

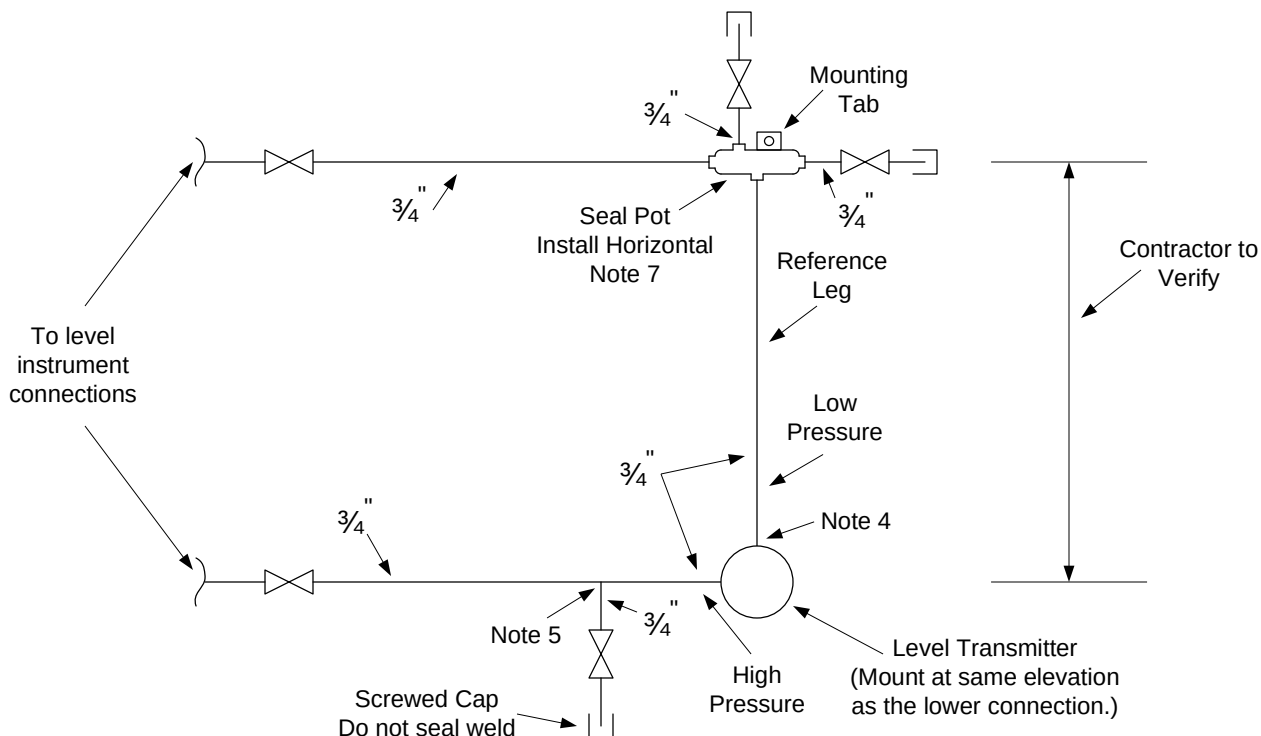
### PROJECT SPECIFICATION

963766 — 604-D

| REV | DATE    | BY  | APVD | REV | DATE | BY | APVD |
|-----|---------|-----|------|-----|------|----|------|
| 0   | 25JUN12 | MVD | MW   |     |      |    |      |
|     |         |     |      |     |      |    |      |
|     |         |     |      |     |      |    |      |
|     |         |     |      |     |      |    |      |

### Sketch " A "

For ASME Flange Class 150, 300 and 600



#### Notes:

1. All piping to be in accordance with UOP Project Specification 801, this project.
2. Contractor shall verify the length of the reference leg to agree with installation for each level transmitter.
3. Reference leg shall be filled with sealing liquid as noted for each transmitter, and transmitter shall be calibrated accordingly.
4. An equalizing valve between the high and low pressure transmitter connections is not permitted.
5. Make connection at the same elevation as the lower vessel nozzle.
6. All valves are gate valves.
7. Install the seal pot at the same elevation as the upper connection. The seal pot shall be at least 3" pipe with a tangent length of at least 6" (150 mm), and metallurgy in accordance with UOP Project Specification 801, this project. The seal pot shall be suitable for the operating conditions, and the pressure rating shall, as a minimum, be stenciled on the shell. Provide a mounting tab to allow for proper support of the seal pot.
8. The use of tubing instead of pipe shall be in accordance with UOP Standard Specification 8-11 Piping. Tubing shall be adequately supported and shall not be used to support the seal pot.

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**posco**  
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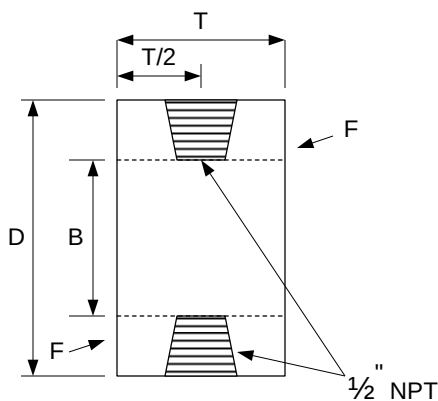
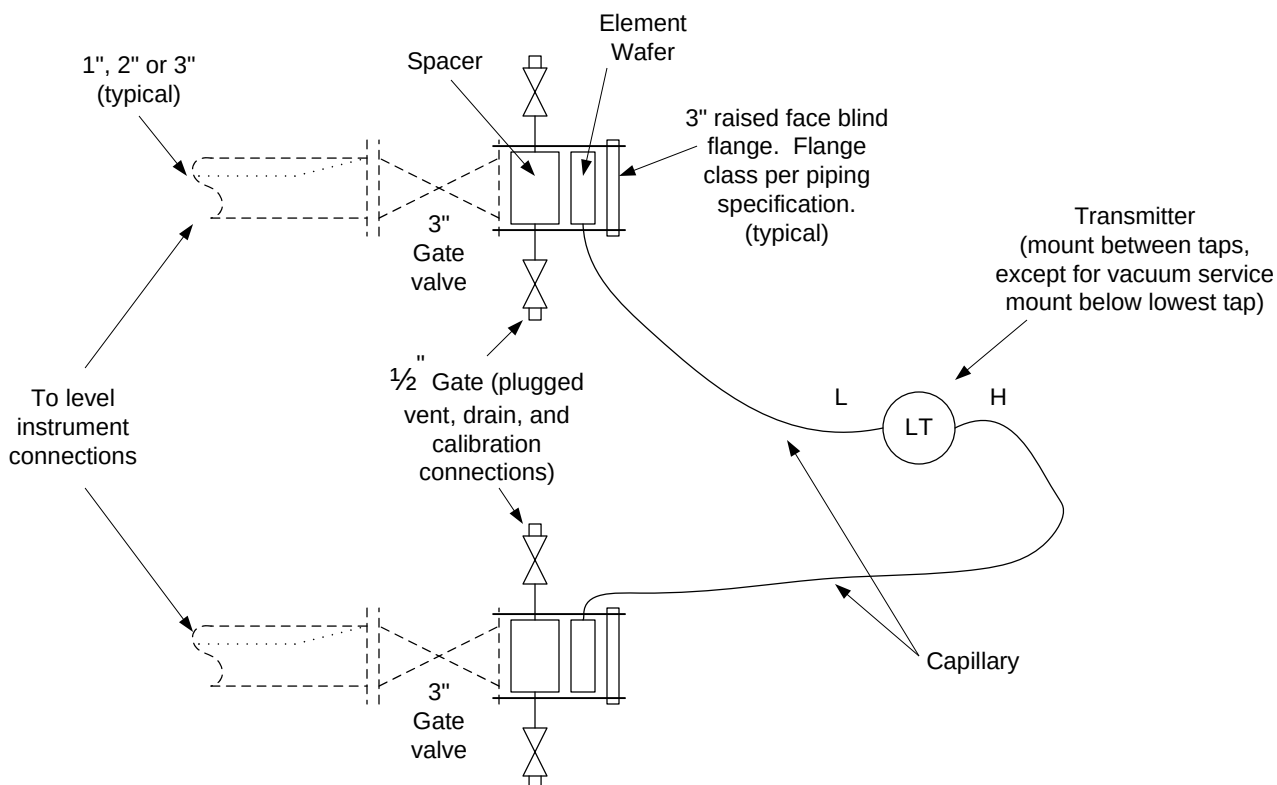
## DIFFERENTIAL PRESSURE INSTRUMENTS

### PROJECT SPECIFICATION

963766 — 604-D

| REV | DATE    | BY  | APVD | REV | DATE | BY | APVD |
|-----|---------|-----|------|-----|------|----|------|
| 0   | 25JUN12 | MVD | MW   |     |      |    |      |
|     |         |     |      |     |      |    |      |
|     |         |     |      |     |      |    |      |
|     |         |     |      |     |      |    |      |

### Sketch " B "



#### SPACER DETAIL (Two required for each instrument)

Material to conform with piping specification for the service.

- T = Thickness of 2" (50 mm)
- D = Outside Diameter (OD of raised facing on flange)
- B = Bore (same as 3" valve flange)
- F = Concentric facing to match flange facing on both sides

#### Notes:

- Heat trace and insulate the nozzle assemblies (including spacer) to maintain the temperature indicated on the Piping and Instrument Diagram. Do not insulate the flange bolts.
- The transmitter and capillaries shall be heat traced and insulated to maintain 150°F (66°C) when the lowest ambient temperature is less than the minimum working temperature of the capillary fill fluid.

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## RADAR LEVEL INSTRUMENTS

### PROJECT SPECIFICATION

963766 — 615

| REV | DATE    | BY  | APVD | REV | DATE | BY | APVD |
|-----|---------|-----|------|-----|------|----|------|
| 0   | 25JUN12 | MVD | MW   |     |      |    |      |
|     |         |     |      |     |      |    |      |
|     |         |     |      |     |      |    |      |
|     |         |     |      |     |      |    |      |

| Item | Quantity | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | 1        | <p>Provide a Time Domain Reflectometry type (Guided Wave) radar level detector and transmitter. The radar instrument must be capable of measuring process fluids with dielectric constants (<math>\epsilon_r</math>) as low as 2.</p> <p><b>Tag Number:</b> LT-23-2001</p> <p><b>Service:</b> Acid Aromatics Sump</p> <p><b>Sensor Data</b></p> <p>Wave Guide Type: Coaxial or Twin rod</p> <p>Flange size, connections: 3" 300 RF</p> <p>Wave Guide Length: by Contractor (based on final dimensions of vessel purchased, vessel drawing shall be provided)</p> <p>Process Wetted Parts: Wave Guide Material – 316/316L SS<br/>O-ring Material – Manufacturer's Recommendation</p> <p><b>Transmitter Data</b></p> <p>Transmitter shall be loop-powered, microprocessor based and shall be capable of being configured locally using a keypad, or remotely using a communicator. Transmitter output shall be 4 to 20 mA. When the transmitter detects an empty vessel condition, the output should be set to zero level value. Local output indicator shall be provided.</p> <p>Response Time: &lt; 1 second maximum response to step change in input</p> <p>Linearity: &lt; 0.1 % of probe length</p> <p>Repeatability: <math>\pm 0.1</math> inch</p> <p>Temp. Effect: <math>\pm 0.01\%</math> of probe length/<math>^{\circ}\text{F}</math> (<math>\pm 0.02\%</math> of probe length/<math>^{\circ}\text{C}</math>)</p> <p><b>Operating Data</b></p> <p>Fluid: Acetone, Phenol, CHP, Cumene, H<sub>2</sub>SO<sub>4</sub> (trace), H<sub>2</sub>O<br/>Presence of solids, acids is possible.<br/>Material is non coating and non bridging.</p> <p>Measurement Length: 2100 mm; 100% level = top of horizontal section</p> <p>Viscosity Range: 0.400 – 4.300 cP</p> <p>Temperature: 38<math>^{\circ}\text{C}</math> , Design Temperature : 325 degC</p> <p>Pressure: 0.01 kg/cm<sup>2</sup>(g) , Design Pressure : 3.5 kg/cm<sup>2</sup>(g)</p> <p><b>Installation</b></p> <p>Installation shall be in accordance with vendor's recommendations. Contractor shall provide vessel drawing and coordinate with vendor to ensure proper installation.</p> <p><b>Supplier (or equal):</b><br/>Magnetrol<br/>5300 Belmont Road<br/>Downers Grove, Illinois 60515-4499 USA<br/>www.magnetrol.com</p> |

 Other specifications  
referenced in this  
specification:

 UOP Standard Specification:  
 UOP Standard Drawing:  
 UOP Project Specification: 601, this project

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## ALARM AND SHUTDOWN SWITCHES

### PROJECT SPECIFICATION

963766 — 644

| REV | DATE    | BY  | APVD | REV | DATE | BY | APVD |
|-----|---------|-----|------|-----|------|----|------|
| 0   | 25JUN12 | MVD | MW   |     |      |    |      |
| 1   | 28AUG12 | MVD | MW   |     |      |    |      |
|     |         |     |      |     |      |    |      |
|     |         |     |      |     |      |    |      |

| Item                                                   | Quantity | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                                                      | 1        | <p>Thermal Dispersion type level switch. The switch relay shall be hermetically sealed SPDT type. Contact rating by Contractor.</p> <p>Available electrical power is 220 VAC, 50 Hz.</p> <p>The switch relay shall deenergize and the contacts shall open when level rises <b>above</b> the switch. The switch contact shall be wired to the control center to actuate an alarm.</p> <p><b>Tag Number:</b> LXSHH -23-0401</p> <p><b>Service:</b> Preflash Column Bottom</p> <p><b>Operating Conditions</b></p> <p>Pressure: -1.0 kg/cm2(g) , Design Pressure : 3.5 kg/cm2(g), F.V</p> <p>Temperature: 70°C , Design Temperature : 250, 195 degC</p> <p>Gas Phase: Cumene</p> <p>Gas Specific Gravity: 4.168</p> <p>Liquid Phase: Hydrocarbon with aromatics</p> <p>Liquid Specific Gravity: 0.916</p> <p>@ Operating Temperature</p> <p><b>Sensor Element:</b></p> <p>Sensor Wetted Parts: 316L stainless steel</p> <p>Sensor Connections: 2 inch 300 RF</p> <p>Sensor Length: By contractor (Nozzle length + immersion)</p> <p>Immersion: Locate tip of sensor approximately 50 mm inside vessel</p> <p>Sensor, sensor extension tube, and flange assembly shall be of welded construction.</p> <p><b>Installation:</b></p> <p>Switch will be installed in a horizontal nozzle on the side of a vessel.</p> <p>Nozzle length: By contractor</p> <p><b>Supplier (or equal):</b></p> <p>Endress + Hauser, Inc.</p> <p>2350 Endress Place</p> <p>PO Box 246</p> <p>Greenwood, Indiana 46143 USA</p> <p>www.endress.com</p> |
| Other specifications referenced in this specification: |          | UOP Standard Specification:<br>UOP Standard Drawing:<br>UOP Project Specification: 601, this project                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                                                                                                                                               |                                                |                           |                                                                                   |              |                                  |                                                   |                                        |     |                       |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|---------------------------|-----------------------------------------------------------------------------------|--------------|----------------------------------|---------------------------------------------------|----------------------------------------|-----|-----------------------|--|
|                                                              |                                                |                           |  |              |                                  | <b>LEVEL INSTRUMENT<br/>PROCESS SPECIFICATION</b> |                                        |     |                       |  |
| CUSTOMERS NAME: PTT PHENOL COMPANY LIMITED                                                                                                    |                                                |                           |                                                                                   |              |                                  | PROJECT No: 120117                                |                                        |     |                       |  |
| LOCATION: MAP TA PHUT, THAILAND                                                                                                               |                                                |                           |                                                                                   |              |                                  | UNIT No: 2300                                     |                                        |     |                       |  |
| REV                                                                                                                                           | O1                                             | A1                        | F1                                                                                | Z1           | DOCUMENT No. 14780-8120-PS-0033  |                                                   |                                        |     |                       |  |
| DATE                                                                                                                                          | DEC 30, 2013                                   | APR 18, 2014              | JUN 25, 2014                                                                      | MAY 10, 2016 |                                  |                                                   |                                        |     |                       |  |
| PREP. BY                                                                                                                                      | S.W.Park                                       | S.W.Park                  | S.W.Park                                                                          | S.W.Park     |                                  |                                                   |                                        |     |                       |  |
| REVIEW BY                                                                                                                                     | J.S.Chung                                      | J.S.Chung                 | J.S.Chung                                                                         | J.S.Chung    |                                  |                                                   |                                        |     |                       |  |
| APP. BY                                                                                                                                       | S.W.Lee                                        | S.W.Lee                   | S.W.Lee                                                                           | S.W.Lee      |                                  |                                                   |                                        |     |                       |  |
| 1                                                                                                                                             | General                                        | Item No.                  | LGR-23-0502                                                                       |              | LGR-23-0602                      |                                                   | LGR-23-0702                            |     | LGR-23-0703           |  |
|                                                                                                                                               |                                                | P&ID No.                  | 14780-8120-25-23-0005A                                                            |              | 14780-8120-25-23-0006            |                                                   | 14780-8120-25-23-0007A                 |     | 14780-8120-25-23-0007 |  |
|                                                                                                                                               |                                                | Service                   | Preflash Column Steam Heater Condensate Drum                                      |              | Concentration Cumene Quench Drum |                                                   | Flash Column Vaporizer Condensate Drum |     | Sample Receiver       |  |
| 3                                                                                                                                             | Location                                       |                           | D-2308                                                                            |              | D-2301                           |                                                   | D-2309                                 |     | Sample Receiver       |  |
| 4                                                                                                                                             | Fluid Properties                               | Type                      |                                                                                   |              |                                  |                                                   |                                        |     |                       |  |
|                                                                                                                                               |                                                | State                     | LIQ                                                                               |              | LIQ                              |                                                   | LIQ                                    |     | LIQ                   |  |
|                                                                                                                                               |                                                | Description               | CONDENSATE                                                                        |              | CUMENE                           |                                                   | CONDENSATE                             |     | HYDROCARBON           |  |
|                                                                                                                                               |                                                | Corrosion/ Erosion due to |                                                                                   |              |                                  |                                                   |                                        |     |                       |  |
| 5                                                                                                                                             |                                                |                           |                                                                                   |              |                                  |                                                   |                                        |     |                       |  |
| LIQUID PROPERTIES                                                                                                                             |                                                |                           |                                                                                   |              |                                  |                                                   |                                        |     |                       |  |
| 10                                                                                                                                            | S.G. @ 15°C                                    |                           |                                                                                   |              |                                  |                                                   |                                        |     |                       |  |
| 11                                                                                                                                            | S.G. @ flow conditions Upper/ Lower            |                           | 0.97                                                                              |              | 0.84                             |                                                   | 0.97                                   |     | 0.9151                |  |
| 15                                                                                                                                            | Liquid Dielectric Constant Upper/ Lower        |                           |                                                                                   |              |                                  |                                                   |                                        |     |                       |  |
| VAPOUR PROPERTIES                                                                                                                             |                                                |                           |                                                                                   |              |                                  |                                                   |                                        |     |                       |  |
| 16                                                                                                                                            | Mol.Wt.                                        |                           |                                                                                   |              |                                  |                                                   |                                        |     |                       |  |
| 17                                                                                                                                            | Vapour Compressibility (PV/RT)                 |                           |                                                                                   |              |                                  |                                                   |                                        |     |                       |  |
| 18                                                                                                                                            | Vapour Density kg/m <sup>3</sup>               |                           |                                                                                   |              |                                  |                                                   |                                        |     |                       |  |
| 21                                                                                                                                            | Operating temp °C                              | Minimum                   |                                                                                   |              |                                  |                                                   |                                        |     |                       |  |
|                                                                                                                                               |                                                | Normal                    | 117                                                                               |              | 38                               |                                                   | 135                                    |     | 70                    |  |
|                                                                                                                                               |                                                | Maximum                   |                                                                                   |              |                                  |                                                   |                                        |     |                       |  |
| 22                                                                                                                                            | Operating Pressure kg/cm <sup>2</sup> g Normal |                           | 0.9                                                                               |              | 0.06                             |                                                   | 2.3                                    |     | -1                    |  |
| 32                                                                                                                                            | Instrument Range (From/ To)                    |                           | (NOTE 1)                                                                          |              | (NOTE 1)                         |                                                   | (NOTE 1)                               |     | (NOTE 1)              |  |
| 33                                                                                                                                            | High Alarm / High High Alarm mm.               |                           | N/A                                                                               |              | N/A                              |                                                   | N/A                                    |     | N/A                   |  |
| 34                                                                                                                                            | Low Alarm / Low Low Alarm mm.                  |                           | N/A                                                                               |              | N/A                              |                                                   | N/A                                    |     | N/A                   |  |
| Reference Point/ Level for 32-34                                                                                                              |                                                | N/A                       |                                                                                   | N/A          |                                  | N/A                                               |                                        | N/A |                       |  |
| 35                                                                                                                                            | Design Pressure kg/cm <sup>2</sup> (G)         |                           | 7                                                                                 |              | 3.5                              |                                                   | 7                                      |     | 10/FV                 |  |
| 36                                                                                                                                            | Design Temperature °C                          |                           | 195                                                                               |              | 120                              |                                                   | 195                                    |     | 250                   |  |
| 37                                                                                                                                            | Remarks                                        |                           | REFLEX TYPE                                                                       |              | REFLEX TYPE<br>NOTE 2            |                                                   | REFLEX TYPE                            |     | REFLEX TYPE           |  |
| 38                                                                                                                                            | Revision                                       |                           |                                                                                   |              |                                  |                                                   |                                        |     |                       |  |
| NOTES<br>1. Refer to gauge glasses data sheet (14780-8550-DS-0008).<br>2. S-12 steam condition (max. 205 C) for steamout shall be considered. |                                                |                           |                                                                                   |              |                                  |                                                   |                                        |     |                       |  |
| <div style="text-align: right;">SHEET 20 OF 22</div>                                                                                          |                                                |                           |                                                                                   |              |                                  |                                                   |                                        |     |                       |  |

|                                                              |                                                             |                                  |  |              | <b>LEVEL INSTRUMENT<br/>PROCESS SPECIFICATION</b> |     |                               |     |                       |     |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|----------------------------------|-----------------------------------------------------------------------------------|--------------|---------------------------------------------------|-----|-------------------------------|-----|-----------------------|-----|--|
| CUSTOMERS NAME: PTT PHENOL COMPANY LIMITED                                                                                                    |                                                             |                                  |                                                                                   |              | PROJECT No: 120117                                |     |                               |     |                       |     |  |
| LOCATION: MAP TA PHUT, THAILAND                                                                                                               |                                                             |                                  |                                                                                   |              | UNIT No: 2300                                     |     |                               |     |                       |     |  |
| REV                                                                                                                                           | O1                                                          | A1                               | F1                                                                                | Z1           | DOCUMENT No. 14780-8120-PS-0033                   |     |                               |     |                       |     |  |
| DATE                                                                                                                                          | DEC 30, 2013                                                | APR 18, 2014                     | JUN 25, 2014                                                                      | MAY 10, 2016 |                                                   |     |                               |     |                       |     |  |
| PREP. BY                                                                                                                                      | S.W.Park                                                    | S.W.Park                         | S.W.Park                                                                          | S.W.Park     |                                                   |     |                               |     |                       |     |  |
| REVIEW BY                                                                                                                                     | J.S.Chung                                                   | J.S.Chung                        | J.S.Chung                                                                         | J.S.Chung    |                                                   |     |                               |     |                       |     |  |
| APP. BY                                                                                                                                       | S.W.Lee                                                     | S.W.Lee                          | S.W.Lee                                                                           | S.W.Lee      |                                                   |     |                               |     |                       |     |  |
| 1                                                                                                                                             | General                                                     | Item No.                         | LGR-23-0804A/B                                                                    |              | LGR-23-1102A/B/C                                  |     | LGR-23-1202                   |     | LGR-23-1304A/B/C      |     |  |
|                                                                                                                                               |                                                             | P&ID No.                         | 14780-8120-25-23-0008A                                                            |              | 14780-8120-25-23-0011                             |     | 14780-8120-25-23-0012A        |     | 14780-8120-25-23-0013 |     |  |
|                                                                                                                                               |                                                             | Service                          | Preflash and Flash Columns<br>Condenser Liquid Boot                               |              | Cumene Flush Drum                                 |     | Decomposer Feed Flush<br>Drum |     | Decomposer Drum       |     |  |
| 3                                                                                                                                             | Location                                                    |                                  | E-2301 BOOTS                                                                      |              | D-2302                                            |     | D-2303                        |     | D-2304                |     |  |
| 4                                                                                                                                             | Fluid<br>Properties                                         | Type                             |                                                                                   |              |                                                   |     |                               |     |                       |     |  |
|                                                                                                                                               |                                                             | State                            | LIQ                                                                               |              | LIQ                                               |     | LIQ                           |     | LIQ                   |     |  |
|                                                                                                                                               |                                                             | Description                      | CUMENE                                                                            |              | HYDROCARBON                                       |     | CUMENE                        |     | HYDROCARBON           |     |  |
|                                                                                                                                               |                                                             | Corrosion/ Erosion due to        |                                                                                   |              |                                                   |     |                               |     |                       |     |  |
| 5                                                                                                                                             |                                                             |                                  |                                                                                   |              |                                                   |     |                               |     |                       |     |  |
|                                                                                                                                               |                                                             | LIQUID PROPERTIES                |                                                                                   |              |                                                   |     |                               |     |                       |     |  |
| 10                                                                                                                                            | S.G. @ 15°C                                                 |                                  |                                                                                   |              |                                                   |     |                               |     |                       |     |  |
| 11                                                                                                                                            | S.G. @ flow conditions    Upper/ Lower                      |                                  | 0.8404                                                                            |              | 0.846                                             |     | 0.846                         |     | 0.919                 |     |  |
| 15                                                                                                                                            | Liquid Dielectric Constant    Upper/ Lower                  |                                  |                                                                                   |              |                                                   |     |                               |     |                       |     |  |
|                                                                                                                                               |                                                             | VAPOUR PROPERTIES                |                                                                                   |              |                                                   |     |                               |     |                       |     |  |
| 16                                                                                                                                            | Mol.Wt.                                                     |                                  |                                                                                   |              |                                                   |     |                               |     |                       |     |  |
| 17                                                                                                                                            | Vapour Compressibility (PV/RT)                              |                                  |                                                                                   |              |                                                   |     |                               |     |                       |     |  |
| 18                                                                                                                                            | Vapour Density                      kg/m <sup>3</sup>       |                                  |                                                                                   |              |                                                   |     |                               |     |                       |     |  |
| 21                                                                                                                                            | Operating<br>temp<br>°C                                     | Minimum                          |                                                                                   |              |                                                   |     |                               |     |                       |     |  |
|                                                                                                                                               |                                                             | Normal                           | 38                                                                                |              | 38                                                |     | 38                            |     | 60                    |     |  |
|                                                                                                                                               |                                                             | Maximum                          |                                                                                   |              |                                                   |     |                               |     |                       |     |  |
| 22                                                                                                                                            | Operating Pressure    kg/cm <sup>2</sup> g    Normal        |                                  | -0.4321                                                                           |              | 0.7                                               |     | 0.7                           |     | 0.7                   |     |  |
| 32                                                                                                                                            | Instrument Range    (From/ To)                              |                                  | (NOTE 1)                                                                          |              | (NOTE 1)                                          |     | (NOTE 1)                      |     | (NOTE 1)              |     |  |
| 33                                                                                                                                            | High Alarm / High High Alarm    mm.                         |                                  | N/A                                                                               |              | N/A                                               |     | N/A                           |     | N/A                   |     |  |
| 34                                                                                                                                            | Low Alarm / Low Low Alarm    mm.                            |                                  | N/A                                                                               |              | N/A                                               |     | N/A                           |     | N/A                   |     |  |
|                                                                                                                                               |                                                             | Reference Point/ Level for 32-34 |                                                                                   | N/A          |                                                   | N/A |                               | N/A |                       | N/A |  |
| 35                                                                                                                                            | Design Pressure                      kg/cm <sup>2</sup> (G) |                                  | 4/FV                                                                              |              | 10                                                |     | 10                            |     | 11                    |     |  |
| 36                                                                                                                                            | Design Temperature                      °C                  |                                  | 250                                                                               |              | 120                                               |     | 120                           |     | 250                   |     |  |
| 37                                                                                                                                            | Remarks                                                     |                                  | REFLEX TYPE                                                                       |              | REFLEX TYPE<br>NOTE 2                             |     | REFLEX TYPE                   |     | REFLEX TYPE           |     |  |
| 38                                                                                                                                            | Revision                                                    |                                  |                                                                                   |              |                                                   |     |                               |     |                       |     |  |
| NOTES<br>1. Refer to gauge glasses data sheet (14780-8550-DS-0008).<br>2. S-12 steam condition (max. 205 C) for steamout shall be considered. |                                                             |                                  |                                                                                   |              |                                                   |     |                               |     |                       |     |  |
| <div style="text-align: right;">SHEET    21    OF    22</div>                                                                                 |                                                             |                                  |                                                                                   |              |                                                   |     |                               |     |                       |     |  |

[illegible]